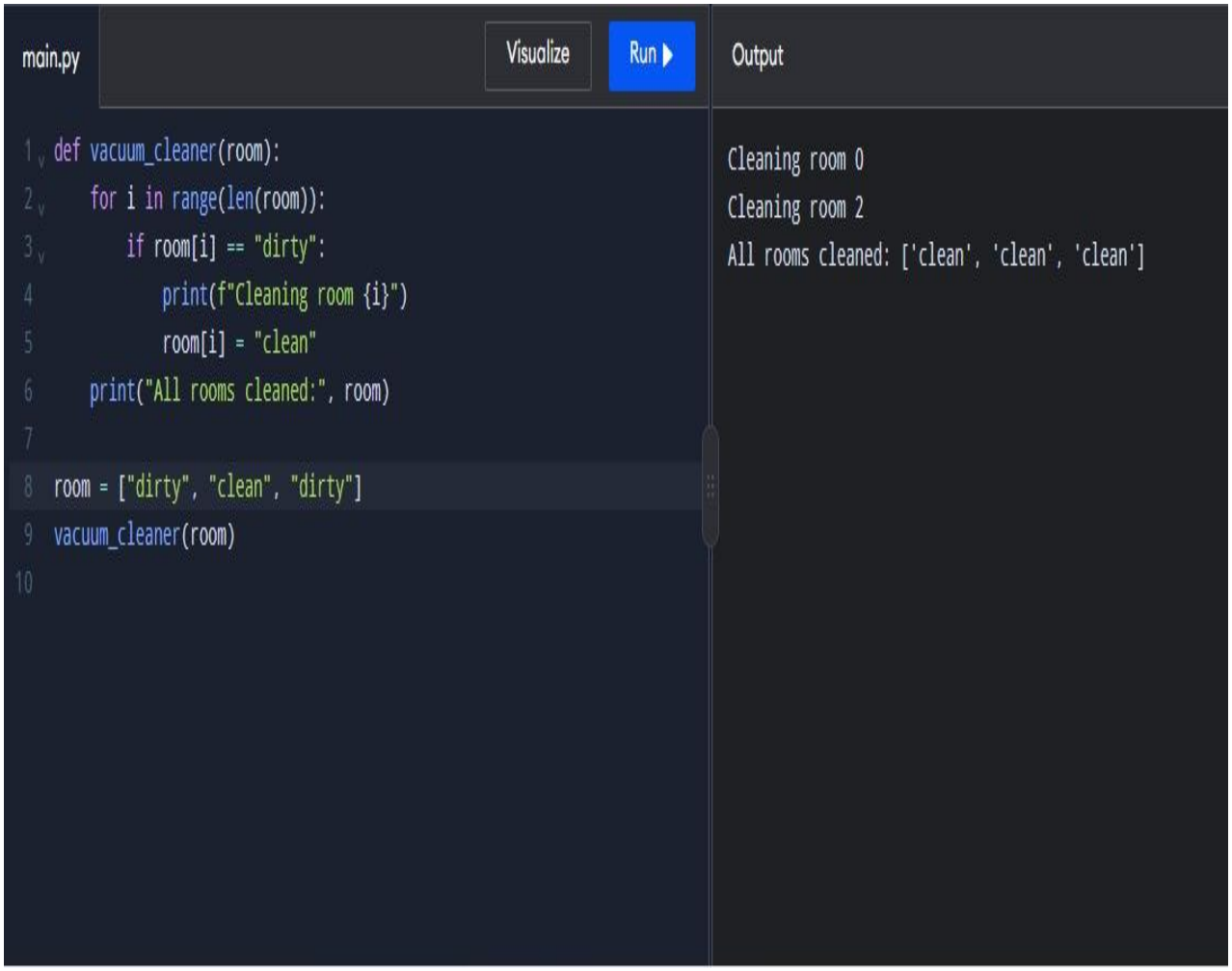
CSA1712 – ARTIFICIAL INTELLIGENCE

EXP 06 : VACUUM CLEANER PROBLEM



EXP 07: BFS

main.py | Visualize | Run ▶

```
from collections import deque
def bfs(graph, start):
    visited = set()
    queue = deque([start])
    while queue:
        node = queue.popleft()
        if node not in visited:
            print(node, end=" ")
            visited.add(node)
            queue.extend(graph[node] - visited)
graph = {
    'A': {'B','C'},
    'B': {'A','D','E'},
    'C': {'A','F'},
    'D': {'B'},
    'E': {'B','F'},
    'F': {'C','E'}
}
bfs(graph, 'A')
```

Output

```
A B C E D F
```

EXP 08: DFS

main.py Visualize Run

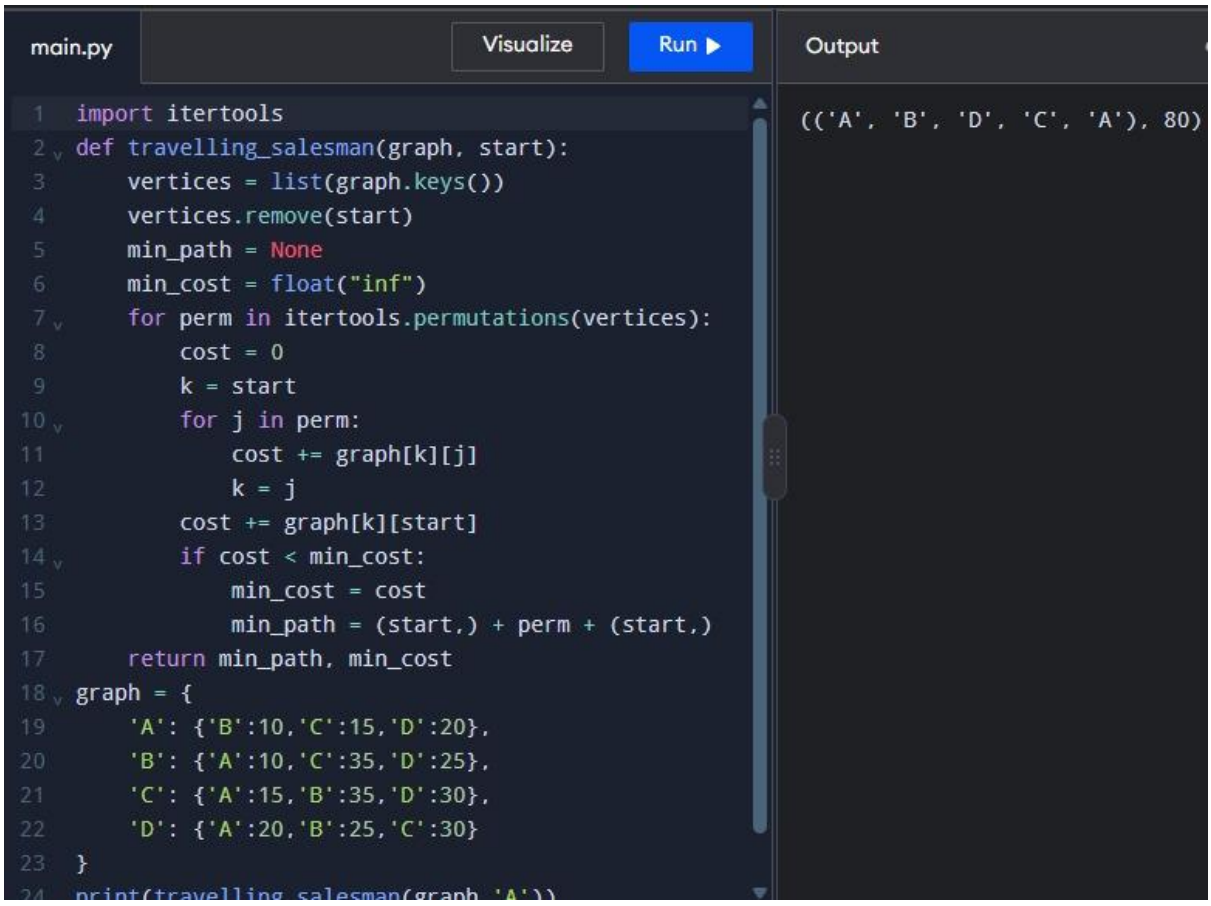
```python
def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()
    visited.add(start)
    print(start, end=" ")
    for neighbor in graph[start]:
        if neighbor not in visited:
    dfs(graph, neighbor, visited)
graph = {
    'A': ['B','C'],
    'B': ['A','D','E'],
    'C': ['A','F'],
    'D': ['B'],
    'E': ['B','F'],
    'F': ['C','E']
}
dfs(graph, 'A')

```

Output

```
A B D E F C
```

EXP 09: TRAVELLING SALESMAN PROBLEM



EXP 10: A* ALGORITHM

